Probability

MATH 356 · Scribed Course Notes · Module 1

Definitions
Examples
Results
Notes

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1 Introduction

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Note 1. The abbreviations used in the manuscript have been expanded in this edition; $AB := A \cap B$, $[n] := \{1, \dots, n\}$ and $(n)_k := n(n-1)\cdots(n-k+1)$ are kept.
Note 2. Red ink in the manuscript (questions, stars) is reproduced in red; blue marginal notes in blue.
Note 3. The handwritten diagrams of the manuscript are redrawn in TikZ; curves that the manuscript only sketches are plotted exactly.
Note 4. Slips of the manuscript are corrected in the text, without comment.

Let us consider the following scenarios.

Example (Roll a fair die)

The face D that shows is a *random variable*.

$$\underbrace{\mathbb{P}}_{\text{probability}}\ (\underbrace{D=2}_{\text{event}}) = \tfrac{1}{6}, \qquad \mathbb{P}(D=k) = \tfrac{1}{6}, \quad k = 1, 2, \dots, 6.$$

□

Example (Roll a fair die twice)

Faces that show $= D_1, D_2$.

$$\mathbb{P}(D_1 = 2,\ D_2 = 3) = \tfrac{1}{6}\cdot\tfrac{1}{6} = \tfrac{1}{36}, \qquad \mathbb{P}(D_1 = k,\ D_2 = j) = \tfrac{1}{6}\cdot\tfrac{1}{6} = \tfrac{1}{36}, \quad j, k = 1, \dots, 6.$$

□

Example (Choose one of two biased dice at random, roll twice)

$$\text{Biased 1: } \begin{cases} 4,5,6 \to \mathbb{P} = \frac{2}{9} \\ 1,2,3 \to \mathbb{P} = \frac{1}{9} \end{cases} \qquad \text{Biased 2: } \begin{cases} 1,2,3 \to \mathbb{P} = \frac{2}{9} \\ 4,5,6 \to \mathbb{P} = \frac{1}{9} \end{cases}$$

$$\begin{aligned} \mathbb{P}(D_1 = 4) &= \tfrac{1}{2}\cdot\tfrac{2}{9} + \tfrac{1}{2}\cdot\tfrac{1}{9} = \tfrac{1}{6}, \\ \mathbb{P}(D_2 = 1) &= \tfrac{1}{2}\cdot\tfrac{1}{9} + \tfrac{1}{2}\cdot\tfrac{2}{9} = \tfrac{1}{6}, \\ \mathbb{P}(D_1 = 1,\ D_2 = 1) &= \tfrac{1}{2}\cdot\tfrac{1}{9}\cdot\tfrac{1}{9} + \tfrac{1}{2}\cdot\tfrac{2}{9}\cdot\tfrac{2}{9} = \tfrac{5}{162}. \end{aligned}$$

The first summand is $\mathbb{P}(\text{chose large die}) \cdot \mathbb{P}(D_1 = 1, D_2 = 1 \mid \text{chose large die})$, where "|" is read "given".

□

Q. What are events? What is independence? What are random variables? How can we model the outcome of an experiment mathematically? How do we do computations with random variables? What are conditional probabilities? What are means / expectations / moments? How do sums of random variables behave? What is universality?

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2 Probability Spaces

Definition

A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$:

- Ω is the *sample space*;
- $\mathcal{F}$ is a set of subsets of Ω, here $\mathcal{F} = 2^\Omega = \mathcal{P}(\Omega)$; elements of $\mathcal{F}$ are called *events*;
- $\mathbb{P}$ is a *probability measure*, i.e. it satisfies *Kolmogorov's axioms*:
  - (i) $\forall E \in \mathcal{F},\ 0 \le \mathbb{P}(E) \le 1$;
  - (ii) $\mathbb{P}(\emptyset) = 0,\ \mathbb{P}(\Omega) = 1$;
  - (iii) for any disjoint events $(E_i)_{i\in\mathbb{N}}$, $\mathbb{P}\Big(\bigsqcup_{i=1}^{\infty} E_i\Big) = \sum_{i=1}^{\infty} \mathbb{P}(E_i)$ (countable additivity).

□

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Exercise. Does (i) follow from (ii) and (iii)?

Example (Roll a fair die once)

$\Omega = \{1, 2, \dots, 6\}$, $\mathcal{F} = 2^\Omega$ = all subsets of Ω, $\mathbb{P}(E) = |E|/6$. For instance $\mathbb{P}(\{1, 3, 5\}) = 3/6 = 1/2$. □

Example (Roll a fair die twice)

Let $[6] := \{1, 2, \dots, 6\}$. Then $\Omega = [6] \times [6]$, $\mathcal{F} = 2^\Omega$, $\mathbb{P}(E) = |E|/36$. □

2.1 Uniform Sampling from a Finite Set

Example (Sampling from $[50] = \{1, \dots, 50\}$)

Say the sample is 7, 12, 43. Questions to think of: with or without replacement? Does the order matter? The answers change the sample space, and with it the probability of the observed sample:

	Ω	$\mathbb{P}((7, 12, 43))$
with replacement, order matters	$\{(a, b, c)\}$	$\frac{1}{\lvert\Omega\rvert} = \frac{1}{50^3}$
w/o replacement, order matters	$\{(a, b, c) :\ a, b, c \text{ distinct}\}$	$\frac{1}{50 \cdot 49 \cdot 48} = \frac{1}{(50)_3}$
w/o replacement, order does not matter	$\{S \subseteq [50] :\ \#S = 3\}$	$\mathbb{P}(\{7, 12, 43\}) = \frac{1}{\binom{50}{3}} = \frac{3!}{(50)_3}$
with replacement, order irrelevant	$\{\{a_1, a_2, a_3\} :\ a_i \in [50]\}$	$\frac{6}{50^3}$

□

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Example (Lotto 6/49)

The probability that the draw has 3 odd and 3 even numbers, computed two ways, in two different sample spaces.

First way: order matters. $\Omega = \{(a,b,c,d,e,f) : \ a,b,\dots,f \in [49] \text{ distinct}\}$, $\mathbb{P}$ = uniform probability measure on Ω, i.e. $\mathbb{P}(E) = |E|/|\Omega|$ for all $E \in \mathcal{F}$. For one fixed pattern, count the tuples that fit it ([49] has 25 odd and 24 even numbers) and divide by $|\Omega| = (49)_6$:

$$\mathbb{P}\big((\text{odd},\text{even},\text{odd},\dots,\text{odd},\text{even})\big) = \frac{(25)_3\,(24)_3}{(49)_6}.$$

Every pattern with 3 odd positions has this same probability, and the patterns are disjoint, so

$$\mathbb{P}(3 \text{ odds, } 3 \text{ evens}) = \underbrace{\binom{6}{3}}_{\text{choosing the odd positions}} \frac{(25)_3\,(24)_3}{(49)_6}.$$

Second way: order does not matter. In a different sample space, $\Omega' = \{S \subseteq [49] : \ \#S = 6\}$ with its uniform probability measure $\mathbb{P}'$, count the sets with 3 odd and 3 even numbers and divide by $|\Omega'| = \binom{49}{6}$:

$$\mathbb{P}'(3 \text{ odds, } 3 \text{ evens}) = \frac{\binom{25}{3}\binom{24}{3}}{\binom{49}{6}}.$$

The two answers agree. Since $(n)_k = k!\binom{n}{k}$,

$$\underbrace{\binom{6}{3}\frac{(25)_3\,(24)_3}{(49)_6}}_{\text{first way, in } \Omega} = \frac{6!}{3!\,3!}\cdot\frac{3!\binom{25}{3}\cdot 3!\binom{24}{3}}{6!\binom{49}{6}} = \underbrace{\frac{\binom{25}{3}\binom{24}{3}}{\binom{49}{6}}}_{\text{second way, in } \Omega'}.$$

□

Remark

Although the counts are different (the first way of computation takes order into consideration, while the second way only chooses), by putting them in their own respective Ω they give rise to the same probability of getting 3 evens and 3 odds. □

2.2 Infinite Probability Spaces

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Example (Toss a coin repeatedly, count the number of tails before the first head)

$\Omega = \{0,1,\dots\}$, $\mathcal{F} = 2^\Omega$.

- Fair coin: $\mathbb{P}(\{k\}) = \mathbb{P}(k \text{ tails before a head}) = \frac{1}{2^k}\cdot\frac{1}{2} = \frac{1}{2^{k+1}}$.
- p-biased coin (probability of head $= p$): $\mathbb{P}(\{k\}) = (1-p)^k\,p$.

Sometimes we refer to this as Geom(p), a *geometric distribution.* □

2.3 Uncountable Probability Spaces

Example (Throw a dart at a uniformly random location on a dart board)

$$\Omega = B(0,1) = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 1\}, \qquad \mathbb{P}(R) := \frac{\text{Area}(R)}{\text{Area}(B(0,1))} \quad \text{for } R \subseteq \Omega.$$

□

Remark

This definition is heuristic. It relies on the measurability of the set R, and the justification of the third of Kolmogorov's axioms relies on the fact that Lebesgue measure satisfies countable additivity. □

Claim

For any point $p \in \Omega$, $\mathbb{P}(\{p\}) = 0$.

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Proof. For any $p \in B(0,1)$, fix ε small; then $\mathbb{P}(B(p,\varepsilon)) = \dfrac{\pi\varepsilon^2}{\pi} = \varepsilon^2$ and $\{p\} \subseteq B(p,\varepsilon) \subseteq B(0,1)$. Thus $\mathbb{P}(\{p\}) \le \varepsilon^2$. Send $\varepsilon \to 0$: $\mathbb{P}(\{p\}) = 0$. □

We used the fact: if $A \subseteq B$, then

$$\mathbb{P}(B) = \mathbb{P}\big(A \cup (B \setminus A)\big) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \ge \mathbb{P}(A).$$

Remark

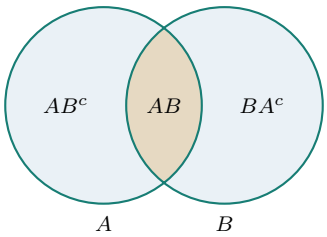
$$\mathbb{P}\Big(\bigcup_{p \in B(0,\frac{1}{2})} \{p\}\Big) = \mathbb{P}\big(B(0,\tfrac{1}{2})\big) = \tfrac{1}{4} \neq \sum_{p \in B(0,\frac{1}{2})} \mathbb{P}(\{p\}) = 0.$$

This is because we are looking at an uncountably infinite sum. □

3 Consequences of the Axioms

3.1 Set-Theoretic Notation vs. Probability

$$\mathbb{P}(A) + \mathbb{P}(A^c) = \mathbb{P}(\Omega) = 1, \quad AB := A \cap B, \quad AB^c = A \setminus B, \quad BA^c = B \setminus A, \quad (AB)^c = A^c \cup B^c.$$



For a sequence of events $E_1, \dots, E_n$,

$$\begin{aligned}\mathbb{P}(E_1 \cup E_2 \cup \cdots \cup E_n) &= \mathbb{P}\big(E_1 \cup (E_1^c E_2) \cup \cdots \cup (E_1^c E_2^c \cdots E_{n-1}^c E_n)\big) \\ &= \mathbb{P}(E_1) + \mathbb{P}(E_1^c E_2) + \cdots + \mathbb{P}(E_1^c \cdots E_{n-1}^c E_n) \\ &\le \mathbb{P}(E_1) + \cdots + \mathbb{P}(E_n).\end{aligned}$$

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Called *subadditivity*.

Principle. To find $\mathbb{P}$ of a complicated event, compute simpler disjoint sub-events, then add up.

Example (Dice game)

Players A, B take turns rolling a fair die, A first. A wins on $\{1,2\}$; B wins on $\{3,4,5,6\}$. Let $A_k = \{A \text{ wins on her } k\text{th roll}\}$.
Want: $\mathbb{P}(A \text{ rolls} \le 4 \text{ times and wins})$. Then $E = A_1 \cup A_2 \cup A_3 \cup A_4$, and

$$\begin{array}{c|cccccccc} A_k: & a_1 & b_1 & a_2 & b_2 & \cdots & a_{k-1} & b_{k-1} & a_k \\ \hline \mathbb{P}: & \frac{2}{3} & \frac{1}{3} & \frac{2}{3} & \frac{1}{3} & \cdots & \frac{2}{3} & \frac{1}{3} & \frac{1}{3} \end{array} \qquad \Longrightarrow \qquad \mathbb{P}(A_k) = \left(\frac{2}{9}\right)^{k-1}\frac{1}{3}.$$

Thus $\mathbb{P}(E) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \mathbb{P}(A_3) + \mathbb{P}(A_4)$, and

$$\mathbb{P}(A \text{ wins}) = \sum_{k=1}^{\infty} \mathbb{P}(A_k) = \sum_{k=1}^{\infty} \left(\frac{2}{9}\right)^{k-1}\frac{1}{3} = \frac{3}{7}.$$

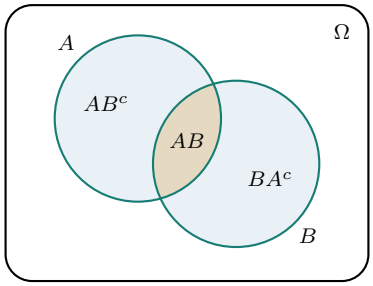
□

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3.2 Complements, Partitions, and Inclusion–Exclusion

Proposition

(a) $\forall A \in \mathcal{F}$, $\mathbb{P}(A) + \mathbb{P}(A^c) = 1$.
(b) $\forall A, B \in \mathcal{F}$, $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$.



Aside on notation.

$$\sum_{i=1}^{n} a_i = \sum_{i \in [n]} a_i, \qquad \sum_{1 \le i < j \le n} a_{\{i,j\}} = \sum_{\{i,j\} \subseteq [n]} a_{\{i,j\}}, \quad a_{ij} = a_{ji} = a_{\{i,j\}},$$

$$\sum_{1 \le i < j < k \le n} a_{\{i,j,k\}} = \sum_{\{i,j,k\} \subseteq [n]} a_{\{i,j,k\}} = \sum_{\substack{S \subseteq [n] \\ |S| = 3}} a_S.$$

These give a more compact way of writing the sum.

Proposition (Inclusion–Exclusion)

For events $E_1, \dots, E_n \subseteq \Omega$,

$$\mathbb{P}\Big(\bigcup_{i\in[n]} E_i\Big) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{\substack{S\subseteq[n]\\|S|=k}} \mathbb{P}\Big(\bigcap_{i\in S} E_i\Big). \tag{$*$}$$

Example

$\mathbb{P}(A\cup B\cup C) = \mathbb{P}(A)+\mathbb{P}(B)+\mathbb{P}(C)-\mathbb{P}(AB)-\mathbb{P}(BC)-\mathbb{P}(AC)+\mathbb{P}(ABC)$. □

Proof of $(*)$. $n=1$, $n=2$ done (geometrically obvious). Suppose the equation holds for $\le n$. We proceed by strong induction. Let $E_1, \dots, E_{n+1} \subseteq \Omega$, $A = E_1 \cup \cdots \cup E_n$, $B = E_{n+1}$. Then

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$$\mathbb{P}(A) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{\substack{S\subseteq[n]\\|S|=k}} \mathbb{P}\Big(\bigcap_{i\in S} E_i\Big), \qquad \mathbb{P}(B) = \mathbb{P}(E_{n+1}),$$

$$\mathbb{P}(A\cap B) = \mathbb{P}\big((E_1\cap E_{n+1})\cup\cdots\cup(E_n\cap E_{n+1})\big) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{\substack{S\subseteq[n]\\|S|=k}} \mathbb{P}\Big(\bigcap_{i\in S} (E_i\cap E_{n+1})\Big).$$

Consider the sum $\mathbb{P}(A\cup B) = \mathbb{P}(A)+\mathbb{P}(B)-\mathbb{P}(A\cap B)$ in cases, comparing the terms with those in $(*)$:

1. if the intersection does not contain E_{n+1}, the term appears in $\mathbb{P}(A)$;
2. if the intersection contains E_{n+1} as the only event, the term is $\mathbb{P}(B)$;
3. if the intersection contains E_{n+1} and other events, the term appears in $\mathbb{P}(A\cap B)$, with sign fixed by "$-\mathbb{P}(A\cap B)$" and a shift of index.

Thus $\mathbb{P}(A\cup B)$ = LHS of $(*)$ = RHS of $(*)$. [By "intersection" we mean a term in the RHS of $(*)$.] □

Example (Hats)

n individuals, n hats. Each individual chooses a hat uniformly at random. What is $\mathbb{P}$(at least one person gets their own hat)?

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Let $E_i := \{\text{person } i \text{ gets their own hat}\}$. Then

$$\mathbb{P}\Big(\bigcup_{i\in[n]} E_i\Big) = \sum_{k\in[n]} (-1)^{k+1} \sum_{\substack{S\subseteq[n]\\|S|=k}} \mathbb{P}\Big(\bigcap_{i\in S} E_i\Big).$$

Furthermore, for a set $\{i_1, \dots, i_k\} \subseteq [n]$, what is $\mathbb{P}(E_{i_1}\cap\cdots\cap E_{i_k})$? Answer: $\dfrac{(n-k)!}{n!}$. Thus

$$\mathbb{P}\Big(\bigcup_{i\in[n]} E_i\Big) = \sum_{k\in[n]} (-1)^{k+1} \sum_{\substack{S\subseteq[n]\\|S|=k}} \frac{(n-k)!}{n!} = \sum_{k\in[n]} (-1)^{k+1} \binom{n}{k} \frac{(n-k)!}{n!}$$

$$= \sum_{k=1}^{n} (-1)^{k+1} \frac{n!}{k!\,(n-k)!} \cdot \frac{(n-k)!}{n!} = -\sum_{k=0}^{n} \frac{(-1)^k}{k!} + 1 \approx 1 - e^{-1}.$$

□

3.3 Continuity of Probability

Reading. Continuity of probability (§1.6). If $A_1 \subseteq A_2 \subseteq \cdots$, then

$$\mathbb{P}\Big(\bigcup_{n\in\mathbb{N}} A_n\Big) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

4 Random Variables

Definition

A *random variable* is a function $X : \Omega \to \mathbb{R}$ for some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. □

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Example

$\Omega = [6]^2$ (two fair die rolls), $X((i,j)) :=$ sum of the two values.

$$\mathbb{P}(X=4) = \mathbb{P}\big(\{(i,j)\in\Omega:\ X((i,j))=4\}\big) = \mathbb{P}\big(\{(i,j)\in[6]^2:\ i+j=4\}\big) = \frac{3}{36} = \frac{1}{12}.$$

□

Example

$\Omega = [6]^{10}$ models rolling a fair die 10 times. $X(\omega) :=$ number of times a 1 is rolled. Then $\mathbb{P}(X=4) = \frac{1}{6^{10}}\binom{10}{4}5^6$. □

Notation. When we say $X \in B$, that means $\text{Range}(X) \subseteq B \subseteq \mathbb{R}$.

Definition

The *distribution* of a random variable X is the collection of probabilities $\mathbb{P}(X \in B)$, $B \subseteq \mathbb{R}$. We may think of this as a function

$$2^{\mathbb{R}} \longrightarrow [0,1], \qquad B \longmapsto \mathbb{P}(X \in B).$$

□

Remark

1. It is in fact impossible to do so in a meaningful way, as we cannot define a measure on $2^{\mathbb{R}}$.
2. We only need $\{\mathbb{P}(X \in (-\infty, b]) :\ b \in \mathbb{R}\}$ to characterize the distribution of a random variable X.

□

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Definition

A random variable X is

- *discrete* if there exists a countable set S such that $\mathbb{P}(X \in S) = 1$;

- *continuous* if there exists $f : \mathbb{R} \to [0, \infty)$ such that for all $a, b \in \mathbb{R}$, $a \leq b$,

$$\mathbb{P}(X \in [a, b]) = \int_a^b f(x)\, dx.$$

For a discrete random variable X taking values in S, the *probability mass function* of X is the function

$$f : S \longrightarrow [0, 1], \qquad s \longmapsto \mathbb{P}(X = s).$$

□

5 Conditioning

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

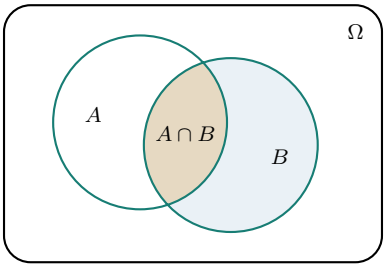
5.1 Conditional Probability and the Multiplication Rule

Definition

Given events A, B with $\mathbb{P}(B) > 0$, let

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

the *conditional probability of A given B*. □



Example (Random sampling of 3 balls without replacement)

Urn with 4 red, 6 purple.

$$\mathbb{P}(2 \text{ red} \mid \geq 1 \text{ red}) = \frac{\mathbb{P}\big((2 \text{ red}) \cap (\geq 1 \text{ red})\big)}{\mathbb{P}(\geq 1 \text{ red})} = \frac{\mathbb{P}(2 \text{ red})}{\mathbb{P}(\geq 1 \text{ red})} = \frac{?/720}{?/720}.$$

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$$\mathbb{P}(2 \text{ red}) = \mathbb{P}(RRP) + \mathbb{P}(RPR) + \mathbb{P}(PRR) = \frac{4 \cdot 3 \cdot 6}{720} \cdot 3,$$
$$\mathbb{P}(\geq 1 \text{ red}) = 1 - \mathbb{P}(PPP) = 1 - \frac{6 \cdot 5 \cdot 4}{720} = \frac{600}{720}.$$

Thus $\mathbb{P}(A \mid B) = \dfrac{3 \cdot 72}{600} = \dfrac{9}{25}$. □

Proposition (Multiplication rule)

$$\mathbb{P}(AB) = \mathbb{P}(A)\,\mathbb{P}(B \mid A) = \mathbb{P}(B)\,\mathbb{P}(A \mid B).$$

More generally, if $\mathbb{P}(A_1 \cdots A_n) > 0$, then

$$\mathbb{P}(A_1 A_2 \cdots A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_3 \mid A_1, A_2) \cdots \mathbb{P}(A_n \mid A_1, \ldots, A_{n-1}).$$

Proof. Exercise. Use the definition of conditional probability and induction on n. □

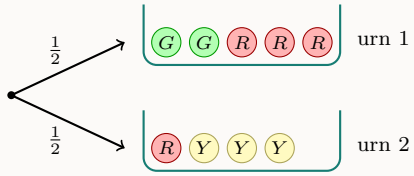
Example (3 draws from the urn)

$$\mathbb{P}(RRP) = \mathbb{P}(\text{1st red}) \cdot \mathbb{P}(\text{2nd red} \mid \text{1st red}) \cdot \mathbb{P}(\text{3rd purple} \mid \text{1st red, 2nd red}) = \frac{4}{10} \cdot \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{10}.$$

□

Example (Two urns)

Urn 1 holds $2G, 3R$ and urn 2 holds $1R, 3Y$; an urn is chosen with probability $\frac{1}{2}$ each, then a ball is drawn.



$$\mathbb{P}(\text{yellow}) = \mathbb{P}(\text{yellow} \cap \text{urn 2}) = \mathbb{P}(\text{urn 2})\,\mathbb{P}(\text{yellow} \mid \text{urn 2}) = \tfrac{3}{8}.$$

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$$\begin{aligned}\mathbb{P}(\text{red}) &= \mathbb{P}\big(\{\text{red}, \text{urn 1}\} \sqcup \{\text{red}, \text{urn 2}\}\big) = \mathbb{P}(\text{red} \cap \text{urn 1}) + \mathbb{P}(\text{red} \cap \text{urn 2}) \\ &= \mathbb{P}(1)\,\mathbb{P}(R \mid 1) + \mathbb{P}(2)\,\mathbb{P}(R \mid 2) = \cdots = \tfrac{17}{40}.\end{aligned}$$

□

5.2 Law of Total Probability and Bayes' Formula

Proposition (Law of total probability)

If events $(B_i)_{i \in [n]}$ partition Ω with $\mathbb{P}(B_i) > 0$ for all $i \in [n]$, then for any event A,

$$\mathbb{P}(A) = \sum_{i \in [n]} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i).$$

Proof. Exercise. □

Proposition (Bayes' formula)

For events A, B with $\mathbb{P}(A), \mathbb{P}(B) \neq 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B \mid A)\,\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\,\mathbb{P}(A^c)}.$$

Example (Base rate fallacy)

The incidence of strep throat in the general population is $\frac{1}{200}$. The rapid strep test has false positive rate $\frac{1}{50}$ and false negative rate $\frac{1}{3}$. Test a random person; $S = \{\text{has strep}\}$, $T = \{\text{tests positive}\}$.

$$\mathbb{P}(S \mid T) = \frac{\mathbb{P}(T \mid S)\,\mathbb{P}(S)}{\mathbb{P}(T \mid S)\,\mathbb{P}(S) + \mathbb{P}(T \mid S^c)\,\mathbb{P}(S^c)} = \frac{\frac{2}{3} \cdot \frac{1}{200}}{\frac{2}{3} \cdot \frac{1}{200} + \frac{1}{50} \cdot \frac{199}{200}} \approx 0.14.$$

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Takeaway. Even if the rapid test is reasonably accurate, whether a positive result implies a high chance of infection depends also on the base rate ($\frac{1}{200}$) of the disease. □

Proposition (General form of Bayes' formula)

If events $\{B_i\}_{i \in [n]}$ partition Ω and $\mathbb{P}(B_i) > 0$ for all i, then for all events A, B with $\mathbb{P}(A), \mathbb{P}(B) > 0$,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\,\mathbb{P}(B)}{\sum_{i \in [n]} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)}.$$

Example (Three types of coins)

Fair coin: $\mathbb{P}(\text{tails}) = \frac{1}{2}$, $\frac{9}{10}$ of coins. Slight bias: $\mathbb{P}(\text{tails}) = \frac{6}{10}$, $\frac{9}{100}$ of coins. Heavy bias: $\mathbb{P}(\text{tails}) = \frac{9}{10}$, $\frac{1}{100}$ of coins.

$$\mathbb{P}(F \mid \text{Tails}) = \frac{\mathbb{P}(\text{Tails} \mid F)\,\mathbb{P}(F)}{\mathbb{P}(\text{Tails} \mid F)\,\mathbb{P}(F) + \mathbb{P}(\text{Tails} \mid S)\,\mathbb{P}(S) + \mathbb{P}(\text{Tails} \mid H)\,\mathbb{P}(H)} = \cdots = \frac{450}{513} \approx 0.88.$$

□

6 Independence

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6.1 Independent Events

Definition

Events A, B are *independent* if $\mathbb{P}(A)\,\mathbb{P}(B) = \mathbb{P}(AB)$. Events $A_1, \ldots, A_n$ are *independent* if for all $I \subseteq [n]$,

$$\mathbb{P}\Big(\bigcap_{i \in I} A_i\Big) = \prod_{i \in I} \mathbb{P}(A_i).$$

□

Example (Two coin tosses)

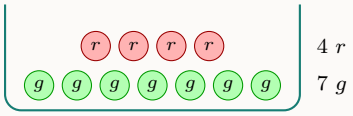
$\Omega = \{T, H\}^2$ with the uniform probability measure, $(\omega_1, \omega_2) \in \Omega$. Let $A = \{\omega_1 = H\}$, $B = \{\omega_2 = H\}$, $C = \{\text{number of heads is even}\}$. Then

$$\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \tfrac{1}{2}, \qquad \mathbb{P}(AB) = \mathbb{P}(BC) = \mathbb{P}(AC) = \tfrac{1}{4}.$$

Yet $\mathbb{P}(ABC) = \mathbb{P}(C \mid AB)\,\mathbb{P}(AB) = 1 \cdot \frac{1}{4} = \frac{1}{4}$. Thus $\{A, B\}$, $\{B, C\}$, $\{A, C\}$ are independent events, while $\{A, B, C\}$ is not. □

Example (Two samples from an urn)

The urn holds 4 red and 7 green balls. $A = \{\text{1st sample red}\}$, $B = \{\text{2nd sample green}\}$.



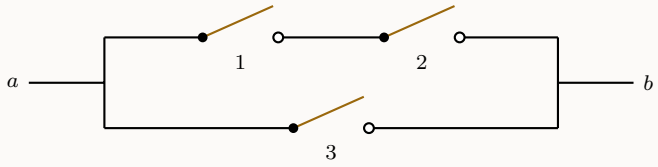
With replacement. $\mathbb{P}(AB) = \dfrac{4 \cdot 7}{121} = \mathbb{P}(A)\,\mathbb{P}(B)$. Hence independent.

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Without replacement. $\mathbb{P}(AB) = \dfrac{4 \cdot 7}{11 \cdot 10} = \dfrac{28}{110}$, while $\mathbb{P}(A) = \dfrac{4}{11}$, $\mathbb{P}(B) = \dfrac{7}{11}$; thus $\mathbb{P}(AB) \neq \mathbb{P}(A)\,\mathbb{P}(B)$, so A, B are not independent. □

Example (Circuit)

Switches 1 and 2 are in series on the top branch, switch 3 alone on the bottom branch, and the two branches are in parallel between a and b.



Let $p_i := \mathbb{P}(\text{switch } i \text{ is on})$; switch states are independent. Then

$$\begin{aligned} \mathbb{P}(\text{current can go } a \to b) &= \mathbb{P}(\text{top}) + \mathbb{P}(\text{bottom}) - \mathbb{P}(\text{both}) \qquad \text{(inclusion–exclusion)} \\ &= p_1 p_2 + p_3 - p_1 p_2 p_3 \qquad \text{by independence.} \end{aligned}$$

□

6.2 Independent Random Variables

Notation.

$$(x_1 + x_3 + x_5)(y_2 + y_6 + y_8)(z_3 + z_6 + z_9) = \sum_{i \in \{1,3,5\}} \sum_{j \in \{2,6,8\}} \sum_{k \in \{3,6,9\}} x_i y_j z_k = \sum_{(i,j,k) \in I \times J \times K} x_i y_j z_k,$$

with $I = \{1, 3, 5\}$, $J = \{2, 6, 8\}$, $K = \{3, 6, 9\}$.

Definition

Random variables $X_1, \dots, X_n : \Omega \to \mathbb{R}$ are (mutually) *independent* if for all $B_1, \dots, B_n \subseteq \mathbb{R}$,

$$\mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n) = \mathbb{P}(X_1 \in B_1) \cdots \mathbb{P}(X_n \in B_n), \quad \text{i.e.} \quad \mathbb{P}\Big(\bigcap_{i \in [n]} \{X_i \in B_i\} \Big) = \prod_{i \in [n]} \mathbb{P}(X_i \in B_i).$$

□

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Recall. A random variable X is discrete if there exists a finite or countable set $S \subseteq \mathbb{R}$ such that $\mathbb{P}(X \in S) = 1$. In this case the set of *possible values* of X is

$$S_X := \{x \in \mathbb{R} \mid \mathbb{P}(X = x) > 0\}.$$

Proposition

Discrete random variables $X_1, \dots, X_n$ are independent if and only if for all possible values x_i of X_i,

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^{n} \mathbb{P}(X_i = x_i).$$

Proof. ($\Rightarrow$) By definition.

($\Leftarrow$) Let $S_i :=$ possible values of X_i. Fix $B_1, \dots, B_n \subseteq \mathbb{R}$. Then

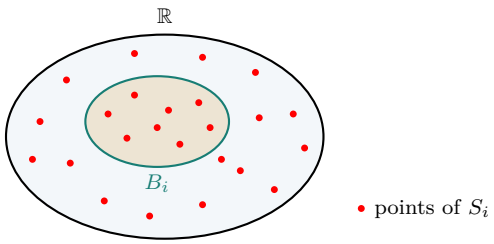
$$\mathbb{P}\Big(\bigcap_{i \in [n]} \{X_i \in B_i\} \Big) = \mathbb{P}\Big(\bigcap_{i \in [n]} \{X_i \in B_i \cap S_i\} \Big)$$

(exercise: show this equality using Kolmogorov's axioms)

$$\begin{aligned}
&= \mathbb{P}\Big(\bigcup_{x_1 \in B_1 \cap S_1} \cdots \bigcup_{x_n \in B_n \cap S_n} \{X_1 = x_1, \dots, X_n = x_n\} \Big) \\
&= \sum_{x_1 \in B_1 \cap S_1} \cdots \sum_{x_n \in B_n \cap S_n} \mathbb{P}(X_1 = x_1, \dots, X_n = x_n) \\
&= \sum_{x_1 \in B_1 \cap S_1} \cdots \sum_{x_n \in B_n \cap S_n} \prod_{i=1}^{n} \mathbb{P}(X_i = x_i) \\
&= \prod_{i=1}^{n} \Big(\sum_{x_i \in B_i \cap S_i} \mathbb{P}(X_i = x_i) \Big) = \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i \cap S_i) \\
&= \prod_{i=1}^{n} \mathbb{P}(X_i \in B_i).
\end{aligned}$$

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So the X_i are independent from the definition. □



Fact

Continuous random variables $X_1, \dots, X_n$ are independent if and only if for all $a_1 < b_1, \dots, a_n < b_n$,

$$\mathbb{P}\big(X_1 \in (a_1, b_1), \dots, X_n \in (a_n, b_n)\big) = \prod_{i=1}^{n} \mathbb{P}\big(X_i \in (a_i, b_i)\big).$$

Example (Sampling k times from $[n]$ with replacement)

$\Omega = [n]^k$ with the uniform probability measure. For $\omega = (\omega_1, \dots, \omega_k) \in [n]^k$ let $X_i(\omega) = \omega_i$. Then

$$\mathbb{P}(X_i = j) = \mathbb{P}(\{\omega : \ \omega_i = j\}) = \frac{n^{k-1}}{n^k} = \frac{1}{n}.$$

To check independence, compute

$$\mathbb{P}(X_1 = j_1, X_2 = j_2, \dots, X_k = j_k) = \mathbb{P}\big(\omega = (j_1, \dots, j_k)\big) = \frac{1}{n^k} = \left(\frac{1}{n}\right)^k = \prod_{i=1}^{k} \mathbb{P}(X_i = j_i).$$

□

Example (Without replacement, previous example)

$\Omega = \{\omega = (\omega_1, \dots, \omega_k) \in [n]^k : \ \omega_i \neq \omega_j \text{ for } i \neq j\}$ with the uniform probability measure, $X_i(\omega) = \omega_i$.

$$\mathbb{P}(X_i = j) = \frac{|\{\omega \in \Omega : \ \omega_i = j\}|}{(n)_k} = \frac{(n-1)_{k-1}}{(n)_k} = \frac{1}{n}.$$

Meanwhile,

$$\mathbb{P}(X_1 = j_1, \dots, X_k = j_k) = \begin{cases} \dfrac{1}{(n)_k}, & \text{if the } j_i \text{ are all distinct,} \\ 0, & \text{otherwise,} \end{cases} \qquad \neq \ \frac{1}{n^k}.$$

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Hence the random variables are not independent. □

6.3 Conditional Independence of Events

Definition

Given events $A, B_1, B_2 \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, we say B_1, B_2 are *conditionally independent given* A if

$$\mathbb{P}(B_1 \cap B_2 \mid A) = \mathbb{P}(B_1 \mid A)\, \mathbb{P}(B_2 \mid A).$$

□

Remark

Conditional independence has **no** implications for independence, and vice versa. □

Example (Counterexample: independent, but not conditionally independent)

Consider X_1, X_2 two independent fair dice outcomes. $B_1 = \{X_1 = 3\}$, $B_2 = \{X_2 = 3\}$, $A = \{X_1 + X_2 = 6\}$.

$$\mathbb{P}(B_1 \mid A) = \frac{\mathbb{P}(B_1 \cap A)}{\mathbb{P}(A)} = \frac{1/36}{5/36} = \frac{1}{5}, \qquad \mathbb{P}(B_2 \mid A) = \frac{\mathbb{P}(B_2 \cap A)}{\mathbb{P}(A)} = \frac{1}{5},$$

$$\mathbb{P}(B_1 \cap B_2 \mid A) = \frac{\mathbb{P}(B_1 \cap B_2 \cap A)}{\mathbb{P}(A)} = \frac{1}{5}.$$

Hence they are independent but not conditionally independent given A. □

Example (Counterexample: conditionally independent, but not independent)

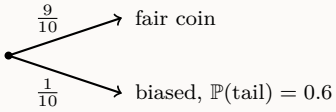
Let X_1, X_2 be samples without replacement from $[10]$. $B_1 = \{X_1 \text{ even}\}$, $B_2 = \{X_2 \text{ even}\}$, $A = \{X_1 \leq 6\} \cap \{X_2 \geq 7\}$.

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B_1 and B_2 are not independent but conditionally independent given A. □

Example (Counterexample: types of coins)

With probability $\frac{9}{10}$ the coin is fair, with probability $\frac{1}{10}$ it is biased with $\mathbb{P}(\text{tail}) = 0.6$.



$B_1 = \{\text{first toss tails}\}$, $B_2 = \{\text{second toss tails}\}$, $A = \{\text{coin fair}\}$.

$$\mathbb{P}(B_1) = \mathbb{P}(B_1 \mid A)\,\mathbb{P}(A) + \mathbb{P}(B_1 \mid A^c)\,\mathbb{P}(A^c) = \tfrac{51}{100}, \qquad \mathbb{P}(B_2) = \tfrac{51}{100},$$

$$\begin{aligned}\mathbb{P}(B_1 \cap B_2) &= \mathbb{P}(B_1 \cap B_2 \mid A)\,\mathbb{P}(A) + \mathbb{P}(B_1 \cap B_2 \mid A^c)\,\mathbb{P}(A^c) \\ &= \tfrac{1}{4} \cdot \tfrac{9}{10} + \left(\tfrac{6}{10}\right)^2 \cdot \tfrac{1}{10} = \tfrac{261}{1000}.\end{aligned}$$

B_1, B_2 are clearly independent if the coin is fair, hence conditionally independent given A, but not independent by the computation above. □

7 Probability Distributions

7.1 Repeated Trials: Bernoulli and Binomial Distributions

Definition

We say a random variable B is $\text{Ber}(p)$-distributed if $\mathbb{P}(B = 1) = p$, $\mathbb{P}(B = 0) = 1 - p$.

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For all $n \in \mathbb{N}$, $p \in [0, 1]$, the *binomial distribution* $\text{Bin}(n, p)$ describes the number of successes among n independent $\text{Ber}(p)$ trials. □

Example (A biased coin with chance p of head)

Then $\Omega = \{0,1\}^n$, $|\Omega| = 2^n$, and for $\omega = (\omega_1, \dots, \omega_n) \in \Omega$,

$$\mathbb{P}(\omega) = p^{\#\{i\in[n]:\ \omega_i=1\}}(1-p)^{\#\{i\in[n]:\ \omega_i=0\}}.$$

Moreover, for the random variable $H : \Omega \to \mathbb{R}$, $\omega \mapsto \#\{i \in [n] :\ \omega_i = 1\} = \|\omega\|_1$,

$$\mathbb{P}(H=k) = \mathbb{P}\big(\{\omega\in\Omega :\ \|\omega\|_1 = k\}\big) = \sum_{\substack{\omega\in\Omega\\ \|\omega\|_1=k}} \mathbb{P}(\omega) = \binom{n}{k} p^k (1-p)^{n-k},$$

so H is $\mathrm{Bin}(n,p)$-distributed by definition. □

Definition

If X is any random variable such that $\mathbb{P}(X=k) = \binom{n}{k}p^k(1-p)^{n-k}$ for $k \in \{0,1,\dots,n\}$, then we say X is $\mathrm{Bin}(n,p)$-distributed. □

7.2 Product Spaces and Infinitely Many Trials

Note (-)

The space $\Omega_i = \{0,1\}$ with $\mathbb{P}_i(0) = 1-p$ is the $\mathrm{Ber}(p)$ probability space, so the space we just described is

$$\Omega_n = \bigotimes_{i=1}^n \Omega_i, \qquad \mathbb{P}_n = \bigotimes_{i=1}^n \mathbb{P}_i \quad \text{(product measure)}.$$

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In particular, restricting Ω_n and $\mathbb{P}_n$ to the first $n-1$ coordinates gives $\Omega_{n-1}, \mathbb{P}_{n-1}$. In fact, one can extend to

$$\Omega_\infty = \bigotimes_{i\geq 1} \Omega_i, \qquad \mathbb{P}_\infty = \bigotimes_{i\geq 1} \mathbb{P}_i,$$

which models an infinite sequence of independent $\mathrm{Ber}(p)$ trials, using the Kolmogorov extension theorem. We take as given that we can define a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which it is possible to define an infinite sequence $(B_i)_{i\geq 1}$ of independent $\mathrm{Ber}(p)$-distributed random variables. □

Observation

For $n \geq 1$, $S := B_1 + \cdots + B_n$ is $\mathrm{Bin}(n,p)$-distributed.

Proof.

$$\mathbb{P}(S=k) = \mathbb{P}\big(\|(B_1,\dots,B_n)\|_1 = k\big) = \mathbb{P}\Big(\bigcup_{\substack{\omega\in\{0,1\}^n\\ \|\omega\|_1=k}} \{(B_1,\dots,B_n)=\omega\}\Big) = \sum_{\substack{\omega\in\{0,1\}^n\\ \|\omega\|_1=k}} \mathbb{P}(B_i = \omega_i\ \forall i)$$

$$\overset{\text{indep.}}{=} \sum_{\substack{\omega\in\{0,1\}^n\\ \|\omega\|_1=k}} \prod_{i=1}^n \mathbb{P}(B_i=\omega_i) = \sum_{\substack{\omega\in\{0,1\}^n\\ \|\omega\|_1=k}} p^k(1-p)^{n-k} = \binom{n}{k} p^k(1-p)^{n-k}.$$ □

7.3 Geometric and Negative Binomial Distributions

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Definition

Say G is Geom(p)-distributed if for $k \in \mathbb{N}$, $\mathbb{P}(G = k) = (1-p)^{k-1}p$. □

Remark

If $N := \inf\{k \geq 1 : \ B_k = 1\}$, then

$$\begin{aligned}\mathbb{P}(N = k) &= \mathbb{P}(B_1 = \cdots = B_{k-1} = 0, \ B_k = 1) \\ &= \mathbb{P}(B_1 = 0)\,\mathbb{P}(B_2 = 0)\cdots\mathbb{P}(B_{k-1} = 0)\,\mathbb{P}(B_k = 1) = (1-p)^{k-1}p.\end{aligned}$$

Hence N is Geom(p)-distributed if the B_i are all identically, independently Ber(p)-distributed. □

Definition

The *negative binomial distribution* NBin(r,p) counts the number of Ber(p) trials until r successes. So Geom(p) = NBin$(1,p)$. □

⋆ Exercise. If X is NBin(r,p)-distributed, what is $\mathbb{P}(X = k)$ for $k \geq r$? Hint: $\mathbb{P}(X = r) = p^r$, $\mathbb{P}(X = r+1) = (1-p)\,p^r \cdot r$.

7.4 The Birthday Problem and the Rayleigh Distribution

Question. Assuming birthdays are uniformly distributed among 365 days, what is the probability of no collision among k individuals?

$$\frac{365}{365}\cdot\frac{364}{365}\cdot\frac{363}{365}\cdots\frac{365-(k-1)}{365}. \qquad \text{? why not } \tfrac{}{365^k} \text{ ?}$$

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Replacing 365 by n,

$$\mathbb{P}(\text{collision free among } k \text{ individuals}) = \prod_{i=1}^{k-1}\Big(1-\frac{i}{n}\Big).$$

Recall, for $\delta > 0$ small, $e^{-\delta} = 1 - \delta + O(\delta^2)$. For small $k \ll n$ and $i \leq k-1$, $1 - \frac{i}{n} \approx \exp(-\frac{i}{n})$, so

$$\mathbb{P}(\text{collision free among } k \text{ individuals}) \approx \prod_{i=1}^{k-1}\exp\Big(-\frac{i}{n}\Big) = \exp\Big(-\sum_{i=1}^{k-1}\frac{i}{n}\Big) = \exp\Big(-\frac{k(k-1)}{2n}\Big).$$

This becomes small when $k \approx \sqrt{n}$, $k = c\sqrt{n}$.

Fix $x > 0$ and let X_n = number of samples until a repeat, when sampling uniformly at random with replacement from $[n]$. Then

$$\mathbb{P}\Big(\frac{X_n}{\sqrt{n}} > x\Big) = \prod_{i=1}^{\lceil x\sqrt{n}\rceil - 1}\Big(1-\frac{i}{n}\Big) \approx \exp\Big(-\frac{\lceil x\sqrt{n}\rceil(\lceil x\sqrt{n}\rceil - 1)}{2n}\Big) \longrightarrow \exp\Big(-\frac{x^2}{2}\Big) \quad \text{as } n \to \infty.$$

Definition

A random variable X is *Rayleigh distributed* if $\mathbb{P}(X > x) = e^{-x^2/2}$ for $x \in \mathbb{R}_{\geq 0}$. □

7.5 Hypergeometric Distribution

N balls, m white, k samples without replacement.

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Definition

The $\mathrm{Hyp}(N, m, k)$ distribution is the distribution of the number of white balls drawn. If N is hypergeometric $\mathrm{Hyp}(N, m, k)$-distributed, then

$$\mathbb{P}(N = j) = \frac{\binom{m}{j}\binom{N-m}{k-j}}{\binom{N}{k}}.$$

□

Note. If instead the sampling is with replacement, then

$$\mathbb{P}(N = j) = \binom{k}{j}\left(\frac{m}{N}\right)^j\left(1 - \frac{m}{N}\right)^{k-j} \sim \mathrm{Bin}\left(k, \frac{m}{N}\right).$$

8 Density and Distribution Functions

8.1 Probability Density Functions

Recall. For a discrete random variable X with possible values $S = \{x \in \mathbb{R} : \mathbb{P}(X = x) > 0\}$, the probability mass function of X is

$$p : S \longrightarrow \mathbb{R}, \qquad x \longmapsto \mathbb{P}(X = x).$$

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Recall. X is *discrete* if its set of possible values is countable, i.e., finite or countably infinite. X is *continuous* if $\exists f : \mathbb{R} \to [0, \infty)$ such that $\forall a \leq b$,

$$\mathbb{P}(X \in [a, b]) = \int_a^b f\,dx.$$

If so,

$$\int_{-\infty}^{\infty} f(x)\,dx = \lim_{n\to\infty}\int_{-n}^{n} f(x)\,dx = \lim_{n\to\infty}\mathbb{P}(X \in [-n, n]) \overset{(*)}{=} 1.$$

Exercise. Formalize this $(*)$ using the definitions. In particular, notice $\mathbb{P}(X \in A) = \mathbb{P}(X^{-1}(A))$.

Notation. If X follows a distribution D, we might write $X \sim D$.

Definition

For X a continuous random variable, if $f : \mathbb{R} \to [0, \infty)$ is such that $\int_a^b f(x)\,dx = \mathbb{P}(X \in [a, b])$, then f is called a *probability density function* for X. □

Remark

A random variable X can have multiple pdfs, but all its pdfs are pairwise different on a null set, i.e., a set of Lebesgue measure 0. Equivalently, if f, g are pdfs of X, then $\forall [a, b]$ we have $\int_a^b f(x)\,dx = \int_a^b g(x)\,dx$. We say in this case that $f = g$ *almost everywhere* (a.e.). □

8.2 Examples of Continuous Random Variables

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Example (Uniform on $[y, z]$)

The only possible (identified mod null sets) pdf is

$$f(x) = \begin{cases} \dfrac{1}{z-y} & \text{if } x \in [y, z], \\ 0 & \text{otherwise.} \end{cases}$$

□

Example (Wigner's semicircle law)

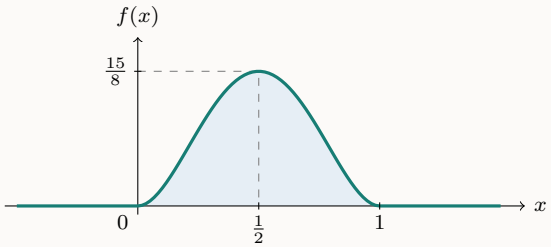
Let $a > 0$. Then

$$f(x) = \begin{cases} \dfrac{2}{\pi a^2}\sqrt{a^2 - x^2}, & \text{if } |x| \le a, \\ 0, & \text{otherwise.} \end{cases}$$

□

Example (Beta$(3, 3)$)

$$f(x) = \begin{cases} 30\, x^2(1-x)^2, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$



For $a, b > 0$, $a, b \in \mathbb{N}$,

$$\text{Beta}(a, b): \qquad f(x) = \begin{cases} \dfrac{\Gamma(a+b)}{\Gamma(a)\,\Gamma(b)}\, x^{a-1}(1-x)^{b-1}, & 0 \le x \le 1, \\ 0, & \text{otherwise,} \end{cases}$$

gamma being the gamma function

$$\Gamma(r) := \int_0^\infty x^{r-1} e^{-x}\, dx.$$

□

Proposition

If X has a continuous pdf f, then for $a \in \mathbb{R}$,

$$f(a) = \lim_{\varepsilon \downarrow 0^+} \frac{\mathbb{P}(X \in [a, a+\varepsilon])}{\varepsilon}.$$

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Proof. Follows directly from the fundamental theorem of calculus and the definition of pdf. □

Remark

The converse holds, i.e., if $\exists X : \Omega \to \mathbb{R}$ a random variable such that

$$f(a) = \lim_{\varepsilon \downarrow 0} \frac{\mathbb{P}(X \in [a, a+\varepsilon])}{\varepsilon},$$

then f is a pdf of X, and X is a continuous random variable. □

Example (Dart board)

$R \to$ uniformly random point on the board. Board $= B(\vec{0}, 9)$. $D \to$ distance from R to $\vec{0}$.
For $0 \le r < 9$, $0 < \varepsilon < 9 - r$,

$$\mathbb{P}(D \in [r, r+\varepsilon]) = \mathbb{P}\Big(R \in \overline{B(\vec{0}, r+\varepsilon)} \setminus \overline{B(\vec{0}, r)}\Big) = \frac{\pi(r+\varepsilon)^2 - \pi r^2}{81\pi} = \frac{2r\varepsilon + \varepsilon^2}{81}.$$

So

$$\lim_{\varepsilon \downarrow 0} \frac{1}{\varepsilon} \mathbb{P}(D \in [r, r+\varepsilon]) = \lim_{\varepsilon \downarrow 0} \frac{2r+\varepsilon}{81} = \frac{2r}{81} \qquad \text{for } 0 \le r < 9.$$

Therefore D is continuous and has density

$$f(x) = \begin{cases} 2x/81, & 0 \le x < 9, \\ 0, & \text{otherwise.} \end{cases}$$

□

8.3 General Random Variables

Generally, a random variable need not be continuous or discrete.

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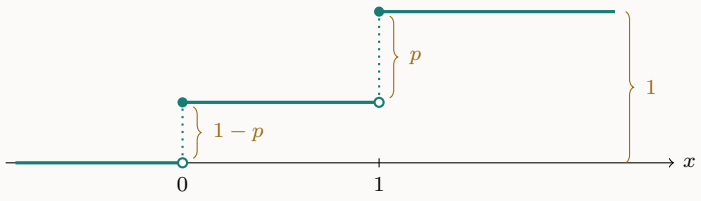
Definition

For a random variable X, its *cumulative distribution function* is the function $F_X : \mathbb{R} \to [0, 1]$ given by

$$F_X(x) = \mathbb{P}(X \le x).$$

□

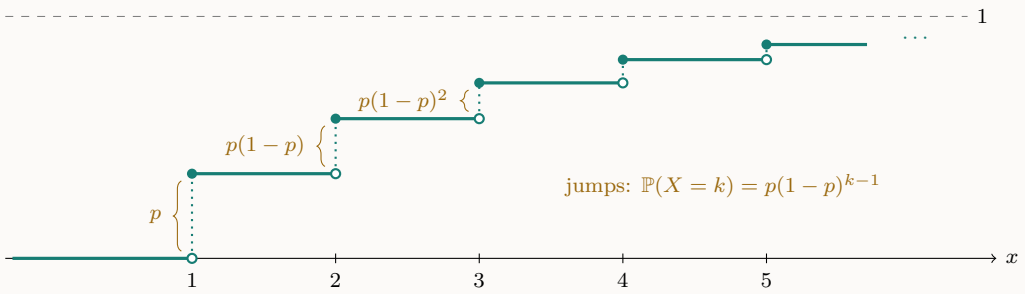
Example ($\mathrm{Ber}(p)$)



$$F(x) = \begin{cases} 0, & x < 0, \\ 1-p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

□

Example (Geom(p))



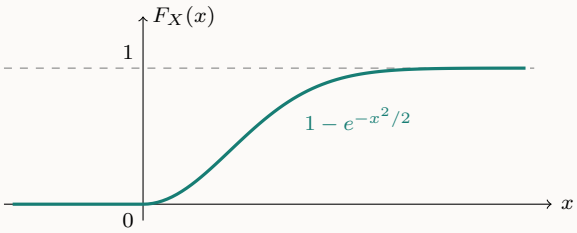
$$F(x) = \mathbb{P}(X \le x) = 1 - \mathbb{P}(X > x) = 1 - (1-p)^{\lfloor x \rfloor},$$

$$F(x) = \begin{cases} 0, & \text{if } x < 1, \\ 1-(1-p)^k & \text{for } k \le x < k+1,\ k \in \mathbb{N}. \end{cases}$$

□

Example (Rayleigh)

$$F_X(x) = \mathbb{P}(X \le x) = 1 - \mathbb{P}(X > x) = 1 - e^{-x^2/2}, \qquad x > 0.$$



□

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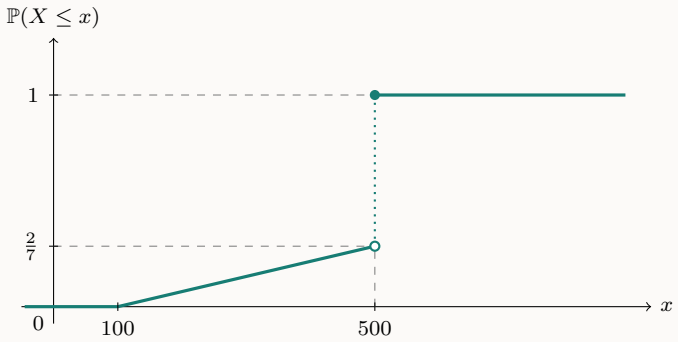
Example (Insurance deductible)

\$500 deductible. Damage to car: $D \sim \text{Unif}[100, 1500]$. Cost to owner: $X = \min(D, 500)$.

$$\mathbb{P}(X \le x) = \mathbb{P}(\min(D, 500) \le x) = \begin{cases} \mathbb{P}(D \le x) & \text{if } x < 500, \\ 1, & x \ge 500. \end{cases}$$

For $100 \le x \le 500$, $\mathbb{P}(D \le x) = \mathbb{P}(D \in [100, x]) = \dfrac{x - 100}{1400}$.

$$\mathbb{P}(X \le x) = \begin{cases} 0, & x < 100, \\ \dfrac{x - 100}{1400}, & 100 \le x < 500, \\ 1, & x \ge 500. \end{cases}$$



□

8.4 Connection between cdf and pdf

For a continuous random variable X with pdf f,

$$\begin{aligned} F(x) &= \mathbb{P}(X \le x) = \mathbb{P}(X \in (-\infty, x]) \\ &= \lim_{n\to\infty} \mathbb{P}(X \in [-n, x]) = \lim_{n\to\infty} \int_{-n}^{x} f(z)\,dz = \int_{-\infty}^{x} f(z)\,dz. \end{aligned}$$

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Conversely, if X has CDF F that is continuous and F' exists except at finitely many points, then

$$f(x) = \begin{cases} F'(x), & F \text{ differentiable at } x, \\ 0, & \text{otherwise} \end{cases}$$

is a pdf for X.

8.5 Connection between cdf and pmf

For a discrete random variable X with possible values S and pmf p_X,

$$F(x) = \mathbb{P}(X \le x) = \mathbb{P}\big(X \in S \cap (-\infty, x]\big) = \sum_{s \in S \cap (-\infty, x]} \mathbb{P}(X = s).$$

In particular, if S has no accumulation (limit) points, then F is piecewise constant.

Exercise. Conversely, if X has cdf F, and F is piecewise constant, then X is discrete, and $S = \{x \in \mathbb{R} : F \text{ is discontinuous at } x\}$.

Proposition (Read proof)

$F : \mathbb{R} \to [0, 1]$ is a cdf $\iff$
① F is nondecreasing;
② F is right continuous, that is, $\lim_{\varepsilon \downarrow 0} F(x + \varepsilon) = F(x)\ \forall x \in \mathbb{R}$;
③ $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$.

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Proposition

If $(E_n)_{n\geq 1}$ are increasing events, then

$$\mathbb{P}\Big(\bigcup E_n\Big) = \lim_{n\to\infty} \mathbb{P}(E_n).$$

If $(F_n)_{n\geq 1}$ are decreasing events, then

$$\mathbb{P}\Big(\bigcap F_n\Big) = \lim_{n\to\infty} \mathbb{P}(F_n).$$

The proposition above is needed to prove the one on the previous page.

Remark (Post-lecture)

1. With the dart example, we worked out that a pdf for the distance to the center of the board was

$$f(x) = \begin{cases} \dfrac{2x}{81}, & 0 \leq x \leq 9, \\ 0, & \text{otherwise.} \end{cases}$$

   With this, we may compute for example

$$\mathbb{P}(1 \leq R \leq 2) = \int_1^2 \frac{2r}{81}\, dr = \frac{3}{81} = \frac{1}{27}.$$

2. If X has cumulative distribution function F, then for any $a < b$, we have

$$\mathbb{P}(X \in (a,b]) = \mathbb{P}\big(\{X \leq b\} \setminus \{X \leq a\}\big) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) = F(b) - F(a).$$

□

9 Expectations

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Expectations = means = averages: $\mathbb{E}X$.

Example

- Fair die roll X: $\frac{1}{6}\cdot 1 + \frac{1}{6}\cdot 2 + \cdots + \frac{1}{6}\cdot 6 = 3.5$.
- Unif$[0,1]$: $X \to \mathbb{E}X = \frac{1}{2}$.
- Unif$[y,z]$: $X \to \mathbb{E}X = \frac{y+z}{2}$.
- Semicircle, $f(x) = \frac{2}{\pi}\sqrt{1-x^2}$ for $-1 \leq x \leq 1$: $X \to \mathbb{E}X = 0$.

□

Definition

For a discrete random variable X, we set

$$\mathbb{E}X = \sum_{s\in S_X} s\cdot \mathbb{P}(X = s) = \sum_{s\in S_X} s\cdot p_X(s),$$

given the sum is finite or the infinite series is absolutely convergent. We require absolute convergence to avoid rearrangements affecting the value of the sum.
For a continuous random variable Y, we set

$$\mathbb{E}Y = \int_{\mathbb{R}} y\, f(y)\, dy,$$

the *expectation* of Y, given $\int_{\mathbb{R}} |y\, f(y)|\, dy < \infty$. □

Example ($X \sim \mathrm{Bin}(n,p)$)

$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, $k \in [n] \cup \{0\}$.

$$\begin{aligned}\mathbb{E}X &= \sum_{k=0}^{n} k\, \mathbb{P}(X=k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} \\ &= \sum_{k=0}^{n} k\, \frac{n!}{k!\,(n-k)!}\, p^k (1-p)^{n-k}\end{aligned}$$

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$$\begin{aligned}&= np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!\,(n-k)!}\, p^{k-1} (1-p)^{n-k} \\ &= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} = np\big(p + (1-p)\big)^{n-1} = np.\end{aligned}$$

□

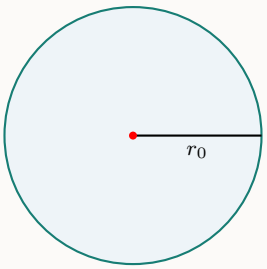
Example ($G \sim \mathrm{Geom}(p)$)

$\mathbb{P}(G = k) = p(1-p)^{k-1}$.

$$\begin{aligned}\mathbb{E}G &= \sum_{k\geq 1} k\, p(1-p)^{k-1} = p \sum_{k\geq 1} \sum_{j=1}^{k} (1-p)^{k-1} \\ &= p \sum_{j\geq 1} \sum_{k\geq j} (1-p)^{k-1} = p \sum_{j\geq 1} \frac{(1-p)^{j-1}}{p} = \frac{1}{p}.\end{aligned}$$

□

Example (Dart board)



D = distance to center,

$$f_D(r) = \frac{2r}{r_0^2}, \qquad 0 \le r \le r_0.$$

$$\mathbb{E}D = \int_0^{r_0} r\, f_D(r)\, dr = \int_0^{r_0} \frac{2r^2}{r_0^2}\, dr = \frac{2}{3}\, r_0.$$

□

Example (Non-example: St. Petersburg paradox)

Casino setup: Pot initially contains \$1.

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Player flips a coin: Heads = win pot; Tail = double value of the pot. $W \to$ winnings.

$$\mathbb{E}W = \sum_{k\ge 0} 2^k\, \mathbb{P}\big((k+1)\text{st toss is first head}\big) = \sum_{k\ge 0} 2^k\, \frac{1}{2^{k+1}} = \infty.$$

Two-player St. Petersburg. First player to flip heads wins 2^k from the other player, where $k = \#$ flips until heads. $Z =$ winnings / loss of player 1.

$$\mathbb{E}Z = \frac{1}{2}\cdot 2^1 - \frac{1}{2^2}\cdot 2^2 + \frac{1}{2^3}\cdot 2^3 - \cdots = 1 - 1 + 1 - 1 + 1 - 1 + \cdots$$

which does not converge. □

9.1 Change of Variables

Example (Fair die roll D)

$$D \le 3 \ \to\ \text{lose \$1}, \qquad D = 4 \ \to\ \text{win \$1}, \qquad D \ge 5 \ \to\ \text{win \$3}.$$

$W =$ winnings.

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$$\mathbb{E}W = -1\cdot\tfrac{1}{6} - 1\cdot\tfrac{1}{6} - 1\cdot\tfrac{1}{6} + 1\cdot\tfrac{1}{6} + 3\cdot\tfrac{1}{6} + 3\cdot\tfrac{1}{6} = \tfrac{2}{3},$$
$$\mathbb{E}W = -1\cdot\tfrac{1}{2} + 1\cdot\tfrac{1}{6} + 3\cdot\tfrac{1}{3} = \tfrac{2}{3}.$$

With the second one, we view W as a new random variable with its own probability mass function. □

Proposition (In general)

If $X : \Omega \to \mathbb{R}$ is discrete and $g : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}(g(X)) = \sum_{s\in S_X} g(s)\cdot\mathbb{P}(X = s) = \sum_{y\in S_{g(X)}} y\cdot\mathbb{P}(g(X) = y),$$

given absolute convergence.

Proof. Exercise. Rearrange terms in the sum. □

Proposition

If X is a random variable with pdf f, and $g : \mathbb{R} \to \mathbb{R}$, then provided absolute integrability,

$$\mathbb{E}(g(X)) = \int_{\mathbb{R}} g(x)\, f(x)\, dx.$$

Proof (guidelines). 1. Prove that if X has density function f with $f(x) = 0$, $x < 0$, then

$$\mathbb{E}X = \int_0^\infty \mathbb{P}(X > x)\, dx.$$

2. Prove with 1 that for $g : \mathbb{R} \to \mathbb{R}_{\geq 0}$,

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$$\mathbb{E}g(X) = \int_0^\infty \mathbb{P}(g(X) > x)\, dx = \int_0^\infty \int_B f_X(y)\, dy\, dx, \qquad B = \{y \in \mathbb{R} \mid g(y) > x\}.$$

3. Change the order of integration via the Fubini–Tonelli theorem,

$$\mathbb{E}g(X) = \int_{\mathbb{R}} \int_0^{g(y)} dx\, f_X(y)\, dy = \int_{\mathbb{R}} g(y)\, f_X(y)\, dy.$$

4. Generalizing this to $g : \mathbb{R} \to \mathbb{R}$, apply what we proved to the positive and negative parts of g:

$$g_+(x) := \begin{cases} g(x), & \text{if } g(x) > 0, \\ 0, & \text{otherwise,} \end{cases} \qquad g_-(x) = \begin{cases} -g(x), & \text{if } g(x) < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Then, by linearity of expectations,

$$\begin{aligned} \mathbb{E}g(X) &= \mathbb{E}g_+(X) - \mathbb{E}g_-(X) \\ &= \int_{\mathbb{R}} g_+(y)\, f_X(y)\, dy - \int_{\mathbb{R}} g_-(y)\, f_X(y)\, dy = \int_{\mathbb{R}} g(y)\, f_X(y)\, dy. \end{aligned}$$

□

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Example (Stick-breaking)

$U \sim \mathrm{Unif}[0, \ell]$, $M = \max(U, \ell - U)$, $g(U) = \max(\ell - U, U)$.

$$\begin{aligned} \mathbb{E}M &= \int_0^\ell \max(x, \ell - x) \cdot f_U(x)\, dx \\ &= \int_0^\ell \max(x, \ell - x) \cdot \frac{1}{\ell}\, dx = 2 \int_{\ell/2}^\ell x \cdot \frac{1}{\ell}\, dx = \cdots \end{aligned}$$

□

Notation. Denote $\sum_{x \in S_X} |x|\, \mathbb{P}(X = x)$ by $\mathbb{E}|X|$, resp. $\int_{-\infty}^\infty |x|\, f_X(x)\, dx$ by $\mathbb{E}|X|$.

Proposition (Linearity of expectation)

① $\mathbb{E}(aX + b) = a\, \mathbb{E}X + b$, if $\mathbb{E}|X| < \infty$.
② Suppose $X, Y : \Omega \to \mathbb{R}$ are random variables with $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$; then

$$\mathbb{E}(X + Y) = \mathbb{E}X + \mathbb{E}Y.$$

Proof. For continuous random variables these are properties of the Lebesgue integral. For discrete random variables, ① is trivial, ② follows by manipulating the definition and rewriting the sums (exercise). □

9.2 Moments

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Definition

For a random variable X, the *nth moment of* X, denoted m_n, is $m_n := \mathbb{E}(X^n)$ whenever $X \geq 0$ or $\mathbb{E}(|X|^n) < \infty$. □

Example ($U \sim \text{Unif}[0,1]$)

$f_U(x) = 1$, $0 \leq x \leq 1$.

$$m_n = \mathbb{E}[U^n] = \int_0^1 x^n \, f_U(x)\, dx \quad \text{(change of variables)} \quad = \int_0^1 x^n \, dx = \frac{1}{n+1}.$$

□

Example ($X \sim \text{Unif}[0,\ell]$)

$f_X(x) = \dfrac{1}{\ell}$, $0 \leq x \leq \ell$. Then $X = \ell \cdot U$. By change of variables,

$$\mathbb{E}(X^n) = \mathbb{E}[(\ell U)^n] = \ell^n \, \mathbb{E}[U^n] = \frac{\ell^n}{n+1}.$$

□

Fact

If X is bounded, i.e. $\text{Range}(X) \subseteq [-M, M]$ for some $M \geq 0$, then the raw moments $(m_n)_{n \in \mathbb{N}}$ determine the distribution of X.

If X is not bounded, this is not true.

Example (Counterexample)

Let

$$\zeta(x) = \sum_{s \geq 1} \frac{1}{s^x}, \qquad x > 1,$$

be the Riemann zeta function. Define a random variable X with

$$\mathbb{P}(X = k) = \frac{1}{\zeta(3)\, k^3}, \qquad k \geq 1.$$

Then

$$\mathbb{E}[X] = \sum_{k \geq 1} k \, \mathbb{P}(X = k) = \sum_{k \geq 1} \frac{1}{\zeta(3)} \cdot \frac{1}{k^2} = \frac{\zeta(2)}{\zeta(3)}, \qquad \mathbb{E}[X^n] = \sum_{k \geq 1} k^n \, \mathbb{P}(X = k) = \infty, \quad n > 1.$$

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Define a random variable Y such that why does such r.v. always exist?

$$\begin{aligned}\mathbb{P}(Y=0) &= \tfrac{1}{2}\,\mathbb{P}(X=1),\\ \mathbb{P}(Y=1) &= 0,\\ \mathbb{P}(Y=2) &= \mathbb{P}(X=2)+\tfrac{1}{2}\,\mathbb{P}(X=1),\\ \mathbb{P}(Y=k) &= \mathbb{P}(X=k) \quad \text{for } k\geq 3.\end{aligned}$$

Then $\mathbb{E}Y = \mathbb{E}X$, $\mathbb{E}(Y^n)=\infty$ for $n>1$, while they have different distributions. □

9.3 Medians and Quantiles

Definition

Given a random variable X, a *median* of X is any value a such that

$$\mathbb{P}(X\leq a)\geq \tfrac{1}{2} \quad \text{and} \quad \mathbb{P}(X\geq a)\geq \tfrac{1}{2}.$$

For $p\in[0,1]$, the *pth quantile* of X is any value a such that $\mathbb{P}(X\leq a)\geq p$ and $\mathbb{P}(X\geq a)\geq 1-p$. □

Example (Dart board)

Dart board of radius r_0, D being the distance to the center. $\mathbb{P}(D\leq r)=r^2/r_0^2$, so a median for D is r such that $r^2/r_0^2=\frac{1}{2}$, so $r=r_0/\sqrt{2}$.
A pth quantile for D is a value r such that $r^2/r_0^2=p$, i.e. $r=r_0\sqrt{p}$. □

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Example ($U\sim \mathrm{Unif}\{1,2,3,\dots,6\}$)

Any value $a\in[3,4]$ satisfies $\mathbb{P}(U\leq a)\geq\frac{1}{2}$, $\mathbb{P}(U\geq a)\geq\frac{1}{2}$. □

Exercise. pth quantile for $p\in(\frac{1}{3},\frac{1}{2})$ with U?

9.4 Variance

Definition

For a random variable X with $\mathbb{E}X=\mu\in\mathbb{R}$, the *variance of* X is

$$\mathrm{Var}(X) := \mathbb{E}\big[(X-\mu)^2\big]\leq\infty.$$

The *standard deviation* of X is $\mathrm{SD}(X)=\sqrt{\mathrm{Var}(X)}=:\sigma$. □

Remark

By linearity of $\mathbb{E}$,

$$\mathrm{Var}(X)=\mathbb{E}\big[(X-\mu)^2\big]=\mathbb{E}[X^2]-2\mu\,\mathbb{E}X+\mu^2=\mathbb{E}[X^2]-\mathbb{E}[X]^2=m_2-m_1^2.$$

□

Example ($X \sim \mathrm{Ber}(p)$)

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X] - \mathbb{E}[X]^2 \quad (X = X^2 \text{ since } X = 0 \text{ or } 1) \quad = p - p^2.$$

□

Example ($U \sim \mathrm{Unif}[a, b]$)

$$\mathbb{E}[U^2] = \int_a^b x^2 \, \frac{1}{b-a} \, dx = \frac{b^2 + a^2 + ab}{3}, \qquad \mathbb{E}[U] = \frac{a+b}{2},$$

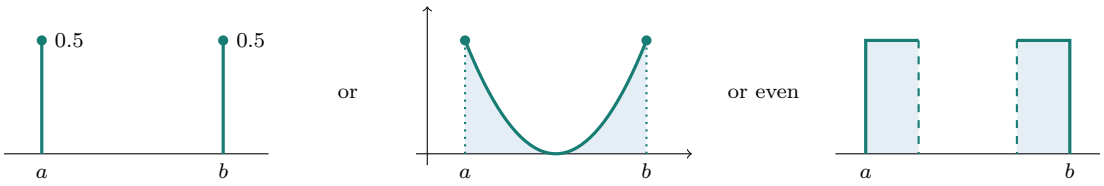
$$\mathrm{Var}(U) = \mathbb{E}[U^2] - \mathbb{E}[U]^2 = \frac{(b-a)^2}{12}.$$

□

Exercise. Come up with a distribution on $[a, b]$ with larger variance than the uniform one.

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Hint.



All these distributions push data further from the mean / $\mathbb{E}$.

Example ($\mathrm{Bin}(n, p)$)

$$\begin{aligned}
\mathbb{E}[X^2] &= \sum_{k=0}^{n} k^2 \, \mathbb{P}(X = k) = np \sum_{k=1}^{n} k \, \frac{(n-1)!}{(k-1)!\,(n-k)!} \, p^{k-1}(1-p)^{n-k} \\
&= np \sum_{k=1}^{n} k \, \mathbb{P}\big(\mathrm{Bin}(n-1, p) = k-1\big) \\
&= np \sum_{k=0}^{n-1} (k+1) \, \mathbb{P}\big(\mathrm{Bin}(n-1, p) = k\big) \\
&= np\Big(\sum k \, \mathbb{P}\big(\mathrm{Bin}(n-1, p) = k\big) + \sum \mathbb{P}\big(\mathrm{Bin}(n-1, p) = k\big) \Big) \\
&= np\big((n-1)p + 1\big) = n^2p^2 - np^2 + np.
\end{aligned}$$

$[\mathbb{E}X]^2 = (np)^2$, so

$$\mathbb{E}[X^2] - (\mathbb{E}X)^2 = np(1-p) = \mathrm{Var}(X).$$

Recall we can view $\mathrm{Bin}(n, p)$ as the sum of n independent Bernoulli distributions, and we observe linearity in variance here:

$$\mathrm{Var}\left(\mathrm{Bin}(n, p)\right) = n \ \mathrm{Var}\left(\mathrm{Ber}(p)\right).$$

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In general, linearity holds for (mutually) independent random variables only. □

Proposition

For a random variable X and $a, b \in \mathbb{R}$,

$$\mathbb{E}[aX + b] = a\,\mathbb{E}X + b, \qquad \mathrm{Var}(aX + b) = a^2\ \mathrm{Var}(X).$$

Proof. We have discussed the linearity of expectation. Variance follows from it:

$$\begin{aligned}\mathrm{Var}(aX+b) &:= \mathbb{E}\Big\{\big[(aX+b) - \mathbb{E}(aX+b)\big]^2\Big\} = \mathbb{E}\Big\{\big[aX + b - a\,\mathbb{E}X - b\big]^2\Big\} \\ &= \mathbb{E}\big[(aX - a\,\mathbb{E}X)^2\big] = a^2\,\mathbb{E}\big[(X - \mathbb{E}X)^2\big] = a^2\ \mathrm{Var}(X).\end{aligned}$$

□

Proposition

$\mathrm{Var}(X) = 0 \iff X$ is *degenerate*, i.e., $\exists a \in \mathbb{R}$ such that $\mathbb{P}(X = a) = 1$.

Proof. ($\Leftarrow$)

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}X)^2\big] = \mathbb{E}\big[(X - a)^2\big] = \mathbb{E}[X^2] - a^2 = a^2 \cdot \mathbb{P}(X = a) - a^2 = a^2 \cdot 1 - a^2 = 0.$$

($\Rightarrow$) For the discrete case, note that

$$\mathrm{Var}(X) = 0 = \sum_{x \in S_X} (x - \mu)^2\,\mathbb{P}(X = x) \iff x = \mu,\ \forall x \in S_X.$$

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Thus X is degenerate by $a = \mu$: $\mathbb{P}(X = \mu) = 1$.

For the continuous case, $\mathrm{Var}(X) > 0$ in fact, and X cannot be degenerate: first notice $\int_{\{c\}} f\,d\lambda = 0$, $\forall c \in \mathbb{R}$, since the singleton set $\{c\}$ has Lebesgue measure 0. Hence by definition, $\mathbb{P}(X = c) = \int_{\{c\}} f\,d\lambda = 0$.

Consider $\mathbb{P}(|X - \mu| < \varepsilon)$ for $\varepsilon > 0$. Then ranging ε over $\frac{1}{n}$, by continuity of measure, $\mathbb{P}(|X - \mu| < \varepsilon) \to \mathbb{P}(X = \mu) = 0$ as $\varepsilon \downarrow 0$. Pick $\varepsilon > 0$ such that $\mathbb{P}(|X - \mu| < \varepsilon) \le \frac{1}{2}$, hence $\mathbb{P}(|X - \mu| \ge \varepsilon) \ge \frac{1}{2}$, and

$$\begin{aligned}\mathrm{Var}(X) &= \int_{\mathbb{R}} (x-\mu)^2 f(x)\,dx \ \ge \int_{|x-\mu|\ge\varepsilon} (x-\mu)^2 f(x)\,dx \ \ge\ \varepsilon^2 \int_{|x-\mu|\ge\varepsilon} f(x)\,dx \\ &= \varepsilon^2\,\mathbb{P}(|X-\mu| \ge \varepsilon) \ \ge\ \frac{\varepsilon^2}{2} > 0.\end{aligned}$$

□

10 The Normal / Gaussian Distribution

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Definition

A random variable X has the *standard Gaussian distribution* if it is continuous and has pdf

$$\varphi(x) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{x^2}{2}}.$$

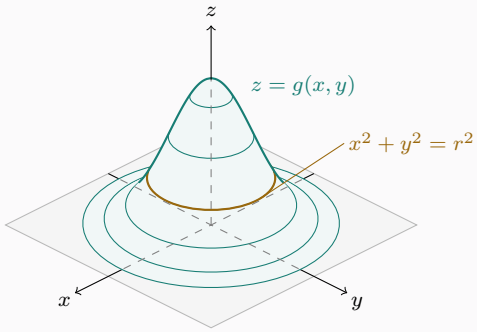
□

Remark

We can then verify that

- $\int_{\mathbb{R}} \varphi(x)\,dx = 1$;

- $\mathbb{E}X = \int_{\mathbb{R}} x\,\varphi(x)\,dx = 0.$



Indeed, define

$$\varphi(x)\varphi(y) =: g(x,y) = \frac{1}{2\pi}\, e^{-\frac{x^2+y^2}{2}},$$

which can be viewed as generated by $\varphi(x)$ rotating along the z-axis, since for every (x,y) such that $x^2+y^2=r^2$, g takes the same value. Thus we can compute

$$\int_{\mathbb{R}}\int_{\mathbb{R}} g(x,y)\,dx\,dy = \int_{\mathbb{R}}\int_{\mathbb{R}} \varphi(x)\varphi(y)\,dx\,dy = \int_{\mathbb{R}} \varphi(x)\,dx \int_{\mathbb{R}} \varphi(y)\,dy = \Big[\int_{\mathbb{R}} \varphi(x)\,dx\Big]^2$$

(by Fubini's theorem), by applying the shell method, viewing the solid generated by revolution of

$$[0,\infty) \to \mathbb{R}, \qquad r \mapsto \frac{1}{2\pi}\, e^{-\frac{r^2}{2}}$$

around the z-axis:

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$$\begin{aligned}\int_{\mathbb{R}}\int_{\mathbb{R}} g(x,y)\,dx\,dy &= \int_0^\infty 2\pi r\cdot\frac{1}{2\pi}\, e^{-\frac{r^2}{2}}\,dr = \int_0^\infty r\, e^{-r^2/2}\,dr = \int_0^\infty e^{-x}\,dx = \int_0^{-\infty} e^{u}\,(-du)\\ &= \int_{-\infty}^0 e^u\,du = e^0 - e^{-\infty} = 1.\end{aligned}$$

Thus $\big(\int_{\mathbb{R}}\varphi(x)\,dx\big)^2 = 1,\ \varphi(x)\ge 0 \implies \int_{\mathbb{R}}\varphi(x)\,dx = 1.$ □

Fact

If X is standard Gaussian, then $\mathbb{E}[X^2] = 1$.

Proof.

$$\begin{aligned}\mathbb{E}[X^2] &= \int_{\mathbb{R}} x^2\varphi(x)\,dx = \int_{\mathbb{R}} \frac{x^2\, e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}\,dx \qquad \big(u = x,\ dv = x\, e^{-\frac{x^2}{2}}\,dx\big)\\ &= \frac{1}{\sqrt{2\pi}}\Big[\big(-x\, e^{-\frac{x^2}{2}}\big)\Big|_{-\infty}^{\infty} + \int_{\mathbb{R}} e^{-\frac{x^2}{2}}\,dx\Big] = \frac{1}{\sqrt{2\pi}}\big[0 - 0 + \sqrt{2\pi}\big] = 1.\end{aligned}$$

□

Since $\mathbb{E}X = 0$, it implies

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2 = 1 - 0^2 = 1.$$

Definition

The standard Gaussian distribution is also called the *standard normal distribution*, denoted by

$$\mathcal{N}\big(\mu = 0, \underbrace{\sigma^2}_{\text{variance}} = 1\big), \qquad \text{short as } \mathcal{N}(0,1).$$

If $X \sim \mathcal{N}(0,1)$, then for $\mu \in \mathbb{R}$, $\sigma > 0$, the random variable $\sigma X + \mu$ is $\mathcal{N}(\mu, \sigma^2)$-*distributed*. □

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Proposition

Suppose $Y = \sigma X + \mu$ is $\mathcal{N}(\mu, \sigma^2)$-distributed, i.e. $X \sim \mathcal{N}(0,1)$. Then

$$\mathbb{E}[\sigma X + \mu] = \sigma\, \mathbb{E}X + \mu = \mu, \qquad \text{and} \qquad \operatorname{Var}[\sigma X + \mu] = \sigma^2 \operatorname{Var}(X) = \sigma^2.$$

Definition

The *cumulative distribution function* of the Gaussian distribution is denoted

$$\Phi(x) := \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}}\, e^{-s^2/2}\, ds.$$

□

Proposition

Furthermore, if Y is $\mathcal{N}(\mu, \sigma^2)$-distributed, then Y has CDF

$$F(y) = \mathbb{P}(Y \le y) = \mathbb{P}(\sigma X + \mu \le y) = \mathbb{P}\Big(X \le \frac{y-\mu}{\sigma}\Big) = \Phi\Big(\frac{y-\mu}{\sigma}\Big)$$

and ("a") pdf

$$f(y) = F'(y) = \frac{d\,\Phi\big(\frac{y-\mu}{\sigma}\big)}{dy} = \frac{1}{\sigma}\,\varphi\Big(\frac{y-\mu}{\sigma}\Big) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(y-\mu)^2}{2\sigma^2}\Big).$$

If $Y \sim \mathcal{N}(\mu, \sigma^2)$, then for $[a,b] \subseteq \mathbb{R}$,

$$\mathbb{P}(a \le Y \le b) = \mathbb{P}(a \le \sigma X + \mu \le b) = \mathbb{P}\Big(\frac{a-\mu}{\sigma} \le X \le \frac{b-\mu}{\sigma}\Big).$$

In particular,

$$\mathbb{P}(|Y - \mu| \le k\sigma) = \mathbb{P}(|X| \le k)$$

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$$= \int_{-k}^{k} \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}\, dx = \Phi(k) - \Phi(-k).$$

$$\begin{aligned} \text{For } k = 1, &\quad \Phi(1) - \Phi(-1) = 0.6826, \\ k = 2, &\quad \Phi(2) - \Phi(-2) = 0.95, \\ k = 3, &\quad \Phi(3) - \Phi(-3) = 0.997, \\ k = 5, &\quad \Phi(5) - \Phi(-5) = 1 - \frac{5.7}{10^7}. \end{aligned}$$

These are known as the *empirical rules*.

End of Midterm 1 Material

11 Selected Problems and Tricks

Worked problems from Homework 1–2 (Anderson–Seppäläinen–Valkó) and Midterm 1, kept here for the *reusable tricks* they carry. Each ends with the trick stated on its own, in the form it will be needed again.

11.1 Hidden Conditioning: the Twins Problem

Example (Homework 1 — twins)

Miranda is expecting twins. A third of twin births are identical twins (always the same sex, that common sex being a fair coin flip); the other two thirds are fraternal twins (two independent fair sexes). (a) What is $\mathbb{P}(\text{both girls})$? (b) Given that both are girls, what is $\mathbb{P}(\text{identical})$?

Solution. The sample space must carry the hidden type: one outcome is a pair $(\text{type}, \text{sexes})$, with masses assigned by the multiplication rule $\mathbb{P}(\text{type})\,\mathbb{P}(\text{sexes} \mid \text{type})$:

$$\mathbb{P}(\text{Id}, GG) = \mathbb{P}(\text{Id}, BB) = \tfrac{1}{3}\cdot\tfrac{1}{2} = \tfrac{1}{6}, \qquad \mathbb{P}(\text{Fr}, **) = \tfrac{2}{3}\cdot\tfrac{1}{4} = \tfrac{1}{6} \ \text{ for each of } GG, GB, BG, BB.$$

Six atoms of mass $\frac{1}{6}$: the problem is secretly a fair die. Then

$$\text{(a)}\quad \mathbb{P}(GG) = \mathbb{P}(\text{Id}, GG) + \mathbb{P}(\text{Fr}, GG) = \tfrac{1}{6} + \tfrac{1}{6} = \tfrac{1}{3}, \qquad \text{(b)}\quad \mathbb{P}(\text{Id} \mid GG) = \frac{1/6}{1/3} = \frac{1}{2}.$$

□

Remark (Trick: model the hidden type)

The tempting answer $\mathbb{P}(GG) = \frac{1}{4}$ assumes the two sexes are independent fair coins — but the premise contradicts this: an identical pair is *one* coin flip copied twice, so same-sex outcomes are inflated ($\mathbb{P}(GG) = \mathbb{P}(BB) = \frac{1}{3}$, $\mathbb{P}(GB) = \mathbb{P}(BG) = \frac{1}{6}$). Whenever a problem mixes types (which coin, which bin, which kind of twin), one outcome of Ω is a complete history *including the type*; probabilities of the visible event are then reassembled by total probability, and conditioning is "cross out the atoms outside B, renormalize by $\mathbb{P}(B)$". Seeing same-sex evidence *does* carry information about the type: $\mathbb{P}(\text{Id} \mid GG) = \frac{1}{2} > \frac{1}{3}$. □

11.2 The k-Selection Problem

Example (Homework 2 — ASV 2.82, the secretary problem)

N candidates arrive one by one in uniformly random order. Strategy: let the first k pass, then hire the first candidate better than everyone seen so far (if none, hire the last). Let $A = \{\text{best candidate hired}\}$. (a) Compute $\mathbb{P}(A)$. (b) Which k is optimal? (c) Find $\lim_{N\to\infty} k^*/N$.

Solution. (a) Condition on the position B of the best candidate; $\mathbb{P}(B = i) = \frac{1}{N}$. For $i \le k$ the best walks away. For $i > k$, the best is hired iff no one in positions $k+1, \dots, i-1$ triggers a hire, which happens iff the best of the first $i-1$ candidates sits among the first k positions; given $B = i$ the remaining relative

order is uniform, so this leader is uniform over $i-1$ slots:

$$\mathbb{P}(A \mid B=i)=\frac{k}{i-1} \qquad \Longrightarrow \qquad P_k := \mathbb{P}(A)=\frac{k}{N}\sum_{j=k}^{N-1}\frac{1}{j} \quad (k\geq 1), \qquad P_0=\tfrac{1}{N}.$$

(b) Compare consecutive values (the *discrete derivative*):

$$P_{k+1}-P_k=\frac{1}{N}\Big(\sum_{j=k+1}^{N-1}\frac{1}{j}-1\Big).$$

The tail is strictly decreasing in k, so (P_k) increases then decreases (unimodal), and

$$k^*=\min\Big\{k\geq 1:\ \sum_{j=k+1}^{N-1}\frac{1}{j}\leq 1\Big\}.$$

(c) Sandwiching the harmonic tail between integrals, $\ln\frac{N}{k^*+1}\leq 1<\ln\frac{N-1}{k^*-1}$, i.e. $\frac{N}{e}-1\leq k^*<\frac{N-1}{e}+1$; dividing by N,

$$\frac{k^*}{N}\longrightarrow\frac{1}{e}\approx 0.368, \qquad \text{and in fact } P_{k^*}\longrightarrow\frac{1}{e} \text{ as well.}$$

□

Remark (Trick: the discrete derivative test)

To maximize a sequence over $k\in\mathbb{Z}$, the analogue of solving $f'=0$ is studying the sign of $\Delta f(k)=f(k+1)-f(k)$; "f' changes sign once" becomes unimodality, and the maximizer is the threshold index. The same move finds the mode of $\mathrm{Bin}(n,p)$ via the ratio $\mathbb{P}(X=k+1)/\mathbb{P}(X=k)$. Division of labour: *integral comparison* locates the maximum only up to $O(\sqrt{N})$ (the top is flat — neighbouring P_k differ by $O(1/N^2)$ while the integral bounds carry $O(1/N)$ slack), so integrals give the asymptotics (c), differences give the exact answer (b). Heuristic first, on scratch paper: maximizing $x\ln(1/x)$ gives $x=1/e$, telling you what to aim for. □

11.3 Von Neumann's Trick: a Fair Coin from a Biased One

Example (Homework 2 — ASV 2.88)

A coin has unknown $\mathbb{P}(H)=p\in(0,1)$. Use it to produce a fair coin.

Solution. Flip twice per round; rounds independent. Within a round,

$$\mathbb{P}(HT)=p(1-p)=\mathbb{P}(TH), \qquad \mathbb{P}(HH)=p^2, \qquad \mathbb{P}(TT)=(1-p)^2.$$

Declare HEADS on HT, TAILS on TH; on HH or TT discard the round and repeat.
Termination. A round is decisive with probability $r=2p(1-p)>0$, so $\mathbb{P}(\text{still running after } m \text{ rounds})=(1-r)^m\to 0$: the procedure stops with probability 1 (the run-forever event is nonempty but null — compare the single point on the dart board).
Fairness. Partition by the first decisive round:

$$\mathbb{P}(\text{HEADS})=\sum_{m\geq 1}(1-r)^{m-1}\,p(1-p)=\frac{p(1-p)}{2p(1-p)}=\frac{1}{2},$$

and identically for TAILS. Equivalently: conditioned on a round being decisive, HT and TH are equally likely. The number of rounds is $\mathrm{Geom}(2p(1-p))$, so the expected number of flips is $\frac{1}{p(1-p)}$ (minimal, 4, at $p=\frac{1}{2}$). □

Remark (Trick: symmetrize so the unknown cancels)

HT and TH have the same probability *whatever* p *is* — the construction manufactures a symmetry instead of estimating p. Two standing sub-patterns: (i) *partition by the first decisive round* (geometric masses summing cleanly — also behind the dice game of §3.1 and the parity computation below); (ii) conditional fairness transfers to unconditional fairness only because the conditioning event has probability one, so the termination half of the argument is load-bearing, not decoration. □

11.4 The Indicator Trick

Example (Homework 2 — ASV 3.78, expected matches)

In the Hats example of §3.2 (n people, uniformly permuted hats), find the expected number of people who receive their own hat.

Solution. Let $E_i = \{\text{person } i \text{ gets their own hat}\}$ and

$$X = \sum_{i=1}^{n} \mathbf{1}_{E_i}, \qquad \mathbb{P}(E_i) = \frac{(n-1)!}{n!} = \frac{1}{n}, \qquad \mathbb{E}[\mathbf{1}_{E_i}] = \mathbb{P}(E_i).$$

By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{P}(E_i) = n \cdot \frac{1}{n} = 1 \qquad \text{for every } n.$$

□

Remark (Trick: unions care about overlap, expectations do not)

The E_i are neither disjoint nor independent ($\mathbb{P}(E_iE_j) = \frac{1}{n(n-1)} \neq \frac{1}{n^2}$) — and linearity does not care. The apparent double-counting is exactly what X measures: a permutation fixing 3 people contributes 3. Contrast with §3.2, where the *probability of the union* $\mathbb{P}(\bigcup E_i) \approx 1 - e^{-1}$ required full inclusion–exclusion. Write the count as a sum of indicators whenever only an expectation is asked; reach for inclusion–exclusion only when a union is asked. □

11.5 A Biased Coin from a Fair One: Binary Expansions

Example (Midterm 1 — the converse of von Neumann)

Given a fair coin and any $p \in (0,1)$ (possibly irrational), produce a $\mathrm{Ber}(p)$ outcome.

Solution. Write $p = 0.p_1p_2p_3\ldots$ in binary. Flip the fair coin to produce independent bits $B_1, B_2, \ldots$, and let $N = \min\{i : B_i \neq p_i\}$ be the first disagreement. Output *success* if $B_N = 0$ (and $p_N = 1$), *failure* if $B_N = 1$.
Termination. Each flip matches p's bit with probability $\frac{1}{2}$ regardless of the bit, so $N \sim \mathrm{Geom}(\frac{1}{2})$: finite almost surely, two flips expected.
Correctness. Success at step i means $B_1 = p_1, \ldots, B_{i-1} = p_{i-1}$, $B_i = 0 < p_i = 1$ — disjoint events, one for each i with $p_i = 1$, of probability 2^{-i}. Hence

$$\mathbb{P}(\text{success}) = \sum_{i:\, p_i = 1} 2^{-i} = p.$$

□

Remark (Trick: realize a number as a disjoint union of events)

The bits define $U = \sum_{i\geq 1} B_i 2^{-i} \sim \mathrm{Unif}[0,1]$, and the procedure outputs $\mathbf{1}_{\{U<p\}}$: the comparison $U < p$ is decided at the first disagreeing bit (lexicographic order), so finitely many flips suffice almost surely; the tie $\{U = p\}$ needs every bit to agree and has probability $\lim_n 2^{-n} = 0$ by continuity of probability. The reusable move runs *backwards* from the usual one: given a series you want to equal a probability (here the expansion of p), hunt for disjoint events carrying exactly those masses — indexed, as usual, by a first decisive time. □

11.6 The Tail-Sum Formula

Proposition (tail-sum formula)

If X takes values in $\{0, 1, 2, \dots\}$, then

$$\mathbb{E}[X] = \sum_{k\geq 1} \mathbb{P}(X \geq k).$$

Continuous twin, for $X \geq 0$ (used in step 1 of the change-of-variables proof, §9.1):

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x)\, dx.$$

Proof. All terms are nonnegative, so the double sum may be reordered:

$$\mathbb{E}[X] = \sum_{k\geq 1} k\,\mathbb{P}(X = k) = \sum_{k\geq 1}\sum_{j=1}^{k} \mathbb{P}(X = k) = \sum_{j\geq 1}\sum_{k\geq j} \mathbb{P}(X = k) = \sum_{j\geq 1} \mathbb{P}(X \geq j).$$ □

Example (minimum and maximum of two dice)

Roll two fair dice independently; let m be the minimum and M the maximum. Tails of minima factor:

$$\mathbb{P}(m \geq k) = \mathbb{P}(\text{both} \geq k) = \left(\frac{7-k}{6}\right)^2 \implies \mathbb{E}[m] = \sum_{k=1}^{6} \left(\frac{7-k}{6}\right)^2 = \frac{36+25+16+9+4+1}{36} = \frac{91}{36}.$$

And since $m + M = D_1 + D_2$, linearity gives $\mathbb{E}[M] = 7 - \frac{91}{36} = \frac{161}{36}$ — no new sums. □

Remark (Trick: compute through tails)

Use the tail-sum when the tail is simpler than the pmf — which is automatic for minima ($\mathbb{P}(\min > k)$ factors over independent coordinates) and for geometrics ($\mathbb{P}(G > k) = (1-p)^k$; this is how $\mathbb{E}[\mathrm{Geom}(p)] = \sum_{k\geq 0}(1-p)^k = \frac{1}{p}$ in one line). Same principle continuously: for $U_1, \dots, U_n$ iid $\mathrm{Unif}[0,1]$, $\mathbb{P}(\min > x) = (1-x)^n$ gives $\mathbb{E}[\min] = \int_0^1 (1-x)^n\, dx = \frac{1}{n+1}$, while $\mathbb{P}(\max \leq x) = x^n$ gives the density nx^{n-1} and $\mathbb{E}[\max] = \frac{n}{n+1}$. □

11.7 First-Step Analysis

Example (gambler's ruin, fair game)

Anna and Boris bet \$1 on independent fair flips until one of them is broke; the total capital is N and Anna currently holds i. What is the probability p_i that Anna ends with everything?

Solution. Condition on the next flip (law of total probability): for $0 < i < N$,

$$p_i = \tfrac{1}{2}\, p_{i-1} + \tfrac{1}{2}\, p_{i+1}, \qquad p_0 = 0,\ p_N = 1.$$

The recursion says $p_{i+1} - p_i = p_i - p_{i-1}$: the increments are constant, so p_i is linear in i, and the boundary values force

$$p_i = \frac{i}{N}.$$

(Smallest case, $N = 3$: $p_1 = \frac{1}{2}p_2$ and $p_2 = \frac{1}{2}p_1 + \frac{1}{2}$ give $p_1 = \frac{1}{3}$, $p_2 = \frac{2}{3}$.) □

Remark (Trick: condition on the first step)

When a random process *restarts* after one step, conditioning on that step converts an unknown probability into a system of linear equations — no series needed. The dice game of §3.1 yields to the same move: $a = \mathbb{P}(A \text{ wins})$ satisfies $a = \frac{1}{3} + \frac{2}{3} \cdot \frac{1}{3}\, a$ (A wins now, or both miss and the game restarts), giving $a = \frac{3}{7}$ without summing the geometric series. □

11.8 Parity via Generating Functions

Example (Homework 2, bonus — $\mathbb{P}(\mathrm{Bin}(n,p)$ is even))

With $q = 1 - p$, the binomial theorem evaluated at $q + p$ and $q - p$ gives

$$1 = (q+p)^n = \sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k}, \qquad (1-2p)^n = (q-p)^n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} p^k q^{n-k}.$$

Adding kills the odd terms and doubles the even ones:

$$\mathbb{P}(X \text{ even}) = \frac{1 + (1-2p)^n}{2}.$$

Sanity: $p = \frac{1}{2}$ gives $\frac{1}{2}$ for all $n \geq 1$; $n = 1$ gives $1 - p$. □

Remark (Trick: evaluate at ± 1 to filter a parity)

Plugging $\pm$ the success weight into the expansion and averaging extracts every second term. The same filtering works by partitioning into consecutive pairs: for $N \sim \mathrm{Geom}(p)$, pairing $\{1,2\}, \{3,4\}, \ldots$ (or summing the even-index geometric series) gives $\mathbb{P}(N \text{ even}) = \frac{1-p}{2-p} < \frac{1}{2}$ — odd indices always carry more mass, since each even index is reached only by surviving the odd one before it. □